

Research

A proposed deep learning model for multichannel ECG noise reduction

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Abstract

Heart disease is a critical concern of healthcare for everyone in today's era. An effective and noninvasive indication of heart disease is an electrocardiogram (ECG). Understanding regular ECG signal patterns and comparisons with irregular, patterns may help to identify the serious nature of heart diseases such as arrhythmia. Comparison of ECG signal patterns is very difficult manually, and machine-based interpretation is a demand of society. Errors in ECG interpretation might result from noise contamination of ECG signals. ECG pretreatment from noise is therefore needed for precise analysis. This article proposed a novel deep learning-based solution for multichannel ECG noise reduction, through utilizing the capabilities of fully convolutional neural network along with the *Jacobin* regularization to ensure confining and preserving local information. Cascaded layered approach was framed in encoder and decoder sections of model where denoising and reconstruction process worked and compared on standard performance parameters with recent denoising deep autoencoders. This proposed work FCN-DAE with Jacobin regularization uses the noise stress test database (NSTDB) for the noise signal of ECG data sourced from the PhysioNet repository. The proposed model achieves $4.763 \times 10^{-2} \text{ mv}^2$ for signal space diversity (SSD), 0.288 mv for the median absolute deviation and 1.859 for the root mean square error (RMSE) in the conducted experiment. The experimental findings demonstrate that complex noise from the ECG signal may be removed up to 97.02%.

Keywords ECG · Deep learning · FCN-DAE · Autoencoder

1 Introduction

Electrocardiography is a technique in which a minor electrical signal is passed to the heart via an electrode, and activity is recorded while the machine provides morphological signal information on the ECG machine. The activity of the heart on the signal parameters is represented and recorded in a medically readable or understandable pattern of PQ and ST segments in ECG signals [1] given in Fig. 1. These segments are reflections of the expansion and contraction of the heart atria and ventricles segmented into four segments: the left atrium, right atrium, left ventricle, right ventricle and (LA, RA, LV, RV). The ordinary function of the heart reflects the patterned wave form due to polarization, depolarization and repolarization of the heart atria and ventricles with specified amplitude levels [3]. This captured waveform ECG machine was analyzed to identify abnormal heart conditions through medical parameters. The accurate analysis of electrocardiogram (ECG) signals plays a pivotal role in diagnosing cardiac abnormalities and monitoring patient health. However, ECG signals are susceptible to various types of noise, such as baseline wandering, muscle artifacts, and electromagnetic

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interference, which can dilute critical information and hinder accurate diagnosis [2]. The guarantee of identifying features present in segments or intervals accurately addresses the challenge of developing robust denoising techniques capable of preserving essential local information within multichannel ECG signals. This local information is generally affected by various types of noise. The brief analysis of the types of noise and their effects on peak, duration and spectral shape is given in Table 1. The noise affects the spatial and local information of the ECG signal that needs to be observed carefully for denoising. To obtain a precise diagnosis, noise must be eliminated from the original ECG signal while preserving as much information as possible. Unfortunately, because all of the noise signals overlap at both low and high frequencies, denoising is a difficult operation. ECG signals have been denoised via a variety of techniques, such as adaptive filtering, wavelet techniques, and empirical mode decomposition, to reduce noise interference [6, 9]. All traditional suggested strategies, however, have high computational and analytical calculation requirements. Additionally, these techniques carry the danger of misinterpretation because cutoff processing can remove clinically significant components of the ECG signal. The denoising of ECGs involves traditional methods to filter noise types and their variations, but in the machine learning arena, autoencoder design via the deep learning method is an effective way to create customized software-based methods for denoising ECGs [16]. Denoising autoencoders (DAEs) are machine learning (ML) technologies that are now available for reconstructing clean data from noisy versions [11]. In recent years, learning approaches, particularly network-based denoising autoencoders, have emerged as powerful tools for denoising ECG signals. These models leverage the hierarchical representation learning capabilities of convolutional neural networks (CNNs) to capture complex temporal and spatial dependencies present in ECG data [4]. By training on pairs of noisy and clean ECG signals, a network-based autoencoder learns to effectively denoise the input signals while preserving crucial local information crucial for accurate diagnosis. Recent research studies in this field have explored various architectures and training strategies to increase the performance of convolutional networks, memory-based long short-term memory (LSTM) models in denoising multichannel ECG signals [12, 39]. The articles presented in past year's related work push and drive gaps for future work. Under the review process, all the articles address the key challenges of conventional filtering methods over the deep learning technique given in Table 2. Deep learning-based autoencoder techniques are popular and helpful for software application design and integration to support the digital healthcare domain. In this view, related work from different articles refers to the category of their addressed issue and solution given in experimental work. Feature engineering is an important challenge in deep learning models for training a model for pattern learning for morphology. Feature engineering addresses improving training accuracy by directing SNR improvements.

1.1 Feature engineering

Arsene [5] demonstrated the effectiveness of combining CNN models with standard filtering methods for denoising ECG signals, which resulted in average $RMS = 0.01419$ and $SNR = 9.6705$ DB over the testing dataset [5]. Finally, Craik [10] provides a comprehensive review of deep learning applications in EEG classification, with a focus on the superior performance of CNNs, RNNs, and DBNs [11]. Franck (2024) addressed the challenge of noise in fetal ECG signals, which may complicate monitoring. This paper reveals that the proposed adversarial deep learning method has a deep convolutional neural network that significantly enhances signal quality through feature engineering, improves the signal-to-noise ratio by 20 dB and increases the detection of QRS complexes while reducing errors [40]. All of the experiments performed here show that feature engineering is a complex phase that needs more layers to be added to network models to perform feature identification. Training from features supported through an autoencoder may reveal differences in standard signal features and noisy signal features.

1.2 Denoising autoencoder

Bouny [8] proposed a convolutional denoising autoencoder (CDAE) specifically for removing additive white Gaussian noise (AWGN) from ECG signals, which resulted in a better signal-to-noise ratio (SNR) and a lower root mean square error (RMSE) than state-of-the-art methods did (0.0113) [8]. Samann (2022) further improved upon this by introducing a multiple parallel hidden layer denoising autoencoder (MPHL-DAE) that achieves better or similar signal-to-noise ratio (SNR) improvements than a multiple hidden layer denoising autoencoder (MHL-DAE) with an SNR of 8.70 dB [29]. Xiong [35, 36] introduced a stacked contractive denoising autoencoder (CDAE) that significantly improved the performance of current denoising autoencoders for ECG signal denoising, with an RMSE of 0.022 and an SNR of 22.75 for 5 dB of noise [35, 36]. These studies collectively highlight the potential of deep learning autoencoders for denoising ECG signals, with each proposing an innovative approach to improve denoising performance. However, there is a gap in the literature in

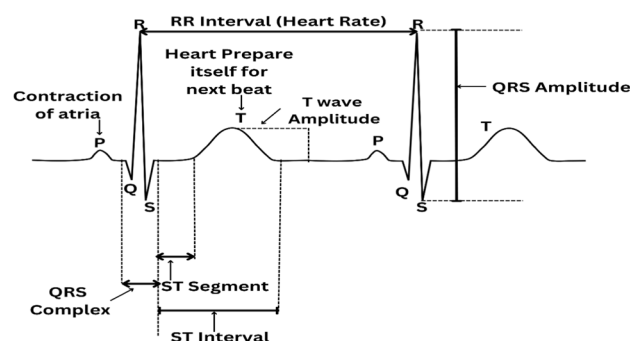
Table 1 Analysis of the types of noise used to collect ECG signals and their effects on local information

Noise Type	Cause	Peak Amplitude	Duration	Spectral Range	Effect on Spatial or Local Information	Limitation
Baseline Wander	Respiration, Electrode Movement	0.1–0.5 mV	Varies	0–0.5 Hz	Distorts baseline, obscures P-QRS-T complexes	Difficult to Consider ST changes in ECG
Powerline Interference	Electrical Equipment, Grid	0.05–0.5 mV	20 ms (60 Hz)	50/60 Hz harmonics	Regular spikes at powerline frequency, obscures signal	Difficult to Consider P & T Peak in ECG
Muscle Artifact	Patient Movement, Electrode Placement	Varies	Varies	20–150 Hz (EMG)	Random spikes, distorts signal during movement	Difficulty to identify R wave (due to low amplitude), and T and P wave (oversensing)
Electromyographic Noise	Muscle Contraction	Varies	Varies	20–150 Hz (EMG)	Irregular spikes, obscures ECG waveform	Difficult to consider R-R interval due to overlapping of QRS spectrum
Electrode Contact Noise	Poor Electrode Contact	Varies	Varies	1 Hz	Abrupt changes, irregular spikes, distortions	Difficulty in collection of ECG (data missing in recording for long collection period)
External Interference	External Electronic Devices	Varies	Varies	Varies	Intermittent spikes, irregular patterns	Direct intrinsic sensing with patient tissue, closed electric circuit is a need

Table 2 Comparison of conventional filters with the deep learning autoencoder

Key Aspect	FCN-DAE Autoencoder	High-Pass/Low-Pass Filtering	Notch Filtering	Wavelet Filtering
Noise Adaptability	Adaptability through automatic training through diverse noise patterns	Adaptable for specific frequency ranges not suitable for nonstationary	Adaptable for fixed-frequency noise, but fails in varying frequencies	Less adaptable due to manual parameter tuning;
Real-Time Capability	Efficient and adaptable for real-time processing	limited by noise assumptions	Limited to static interference	Computationally intensive
Handling Mixed Noise	Robust due to its data-driven training	not suitable (baseline wander + muscle artifacts)	Not suitable	Moderate effectiveness but dependent on parameters
Generalization	Excellent generalization due to diverse noise learned patterns	Poor generalization	Effective for predefined interference	Moderate generalization and depends on wavelet choice
Preservation of ECG Features	Best at maintaining ECG morphology	May distort low-amplitude features like P-waves	May remove overlapping signaling components	Moderate feature preservation
Parameter Tuning	automatically through learning of optimal features	manual cutoff frequency	Exact noise frequency	Selection of wavelet basis and thresholds
Implementation	Requires initial training but straightforward application	Simple but inflexible	Simple for specific tasks	Complex and requires domain expertise
Signal Preservation	Retains ECG morphology more effectively, preserving P, QRS, and T waves	May distort or remove low/high-frequency components of the ECG signal	May distort ECG if noise frequency overlaps with signaling components	Can overfilter or underfilter depending on threshold
Quantitative Performance	Consistently high SNR, low Percentage Root Mean Square Difference (PRD)	Moderate Signal-to-Noise Ratio (SNR); risk of residual noise	High SNR for targeted interference	Good SNR with proper tuning but prone to over/underfiltering

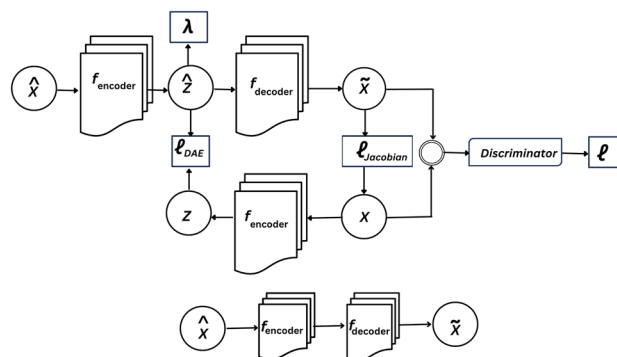
Fig. 1 Standard ECG signal showing local information segments and intervals of the respective peaks



terms of a comprehensive comparison of these different deep learning autoencoder models for denoising ECG signals. Further research is needed to determine the most effective model for this specific application on the basis of the optimizer and objective function design. Chatterjee [9] further discussed the use of an FCN-based autoencoder in denoising ECG signals, emphasizing its effectiveness in removing various types of noise [9]. Kim [20] underscores the value of denoising autoencoders in handling missing data in healthcare, suggesting their potential in improving the performance of subsequent classifiers. These studies collectively support the use of the FCN-DAE in denoising ECG signals, particularly in the context of healthcare applications, with a model for denoising ECG signal waveforms that preserves local information, which is necessary for early arrhythmia detection. In most of the related article author used dilated convolutions to capture long-range dependencies across different leads while maintaining computational efficiency [13]. Additionally, these attention-based mechanisms have been employed to concentrate on informative regions of the signal, such as the QRS complex, thereby improving denoising performance [15]. Despite this attention being given to informative regions, different challenges persist in ECG denoising [14]. The key challenges that were raised from previous studies, preserving morphological information within signal after denoising and second is to build a robust learning model for multichannel ECG, which can perform well on feature engineering stage or achieve significant performance at reconstruction stage. Using proposed deep learning model introduces a novel FCN-based denoising autoencoder specifically optimized for multichannel ECG signals. A key point of innovation is to build robust autoencoder and the integration of a Jacobian function that helps to improve feature sensitivity and preserve critical morphology in denoised signals. The Jacobian function plays a crucial role in ensuring the stability and effectiveness of the learning network for classification task. The Jacobian matrix captures the local sensitivity of the model's outputs with respect to its inputs, which is essential for analyzing noise propagation through the deep learning layers. The Jacobian function helps regulate the transformation of input signals, ensuring that small perturbations (such as noise) do not lead to significant distortions in the output. By incorporating the Jacobian function/ regularization into our proposed model, we enhance model robustness, improve generalization, and prevent excessive signal smoothing, which may otherwise remove vital ECG characteristics. Additionally, using the Jacobian function allows for better gradient flow analysis, ensuring that the deep learning model does not suffer from vanishing or exploding gradients, which can hinder training convergence. This contributes to stable learning and improved noise reduction performance across multiple ECG channels. This integration of the Jacobian function into our proposed model practically beneficial for achieving high-fidelity ECG signal reconstruction while effectively suppressing noise. This proposed work outperforms conventional methods DRNN by reducing different types of ECG noise and maintaining computational efficiency along with clinical interpretability.

2 Material and Methods

This research article proposes a deep learning model for confining and preserving local information from multichannel noise in electrocardiogram (ECG) signals via a fully convolutional-based network shown in Fig. 2. Leveraging the hierarchical representation learning capabilities of convolutional neural networks, our proposed fully convolutional network-based denoising autoencoder (FCN-DAE) architecture incorporates dilated convolutions to capture long-range dependencies within each channel while preserving spatial information across multiple leads. By training the model on pairs of noisy and clean ECG signals, our objective function (Jacobian matrix-based) balances noise reduction with the preservation of essential local features, such as the shape of the QRS wave. The proposed work is carried out by implementing two models, (1) a deep recurrent neural network (DRNN) and (2) an FCN- DAE, and the results are compared

Fig. 2 Proposed Model

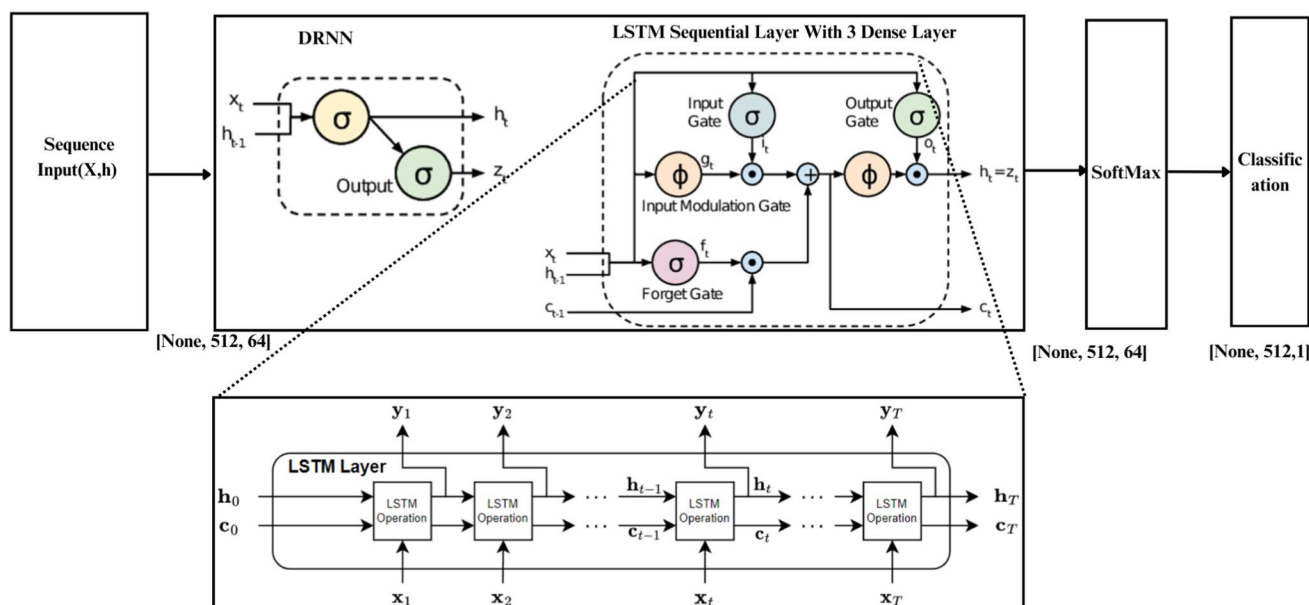
for multichannel noise reduction in an ECG signal. In both models, the initial configuration of the layer was chosen from empirical experiments in the literature, and the configuration of the number of layers was customized to find the best configuration for the experiment.

2.1 DRNN

The DRNN is an advanced neural network that extends the recurrent neural network (RNN) through its layer stacking. Stacking in layers helps in identifying and capturing long-term and short-term dependencies. Each unit in an LSTM layer processes one element of a sequence at a time while maintaining an internal "memory" to track information over time. Our experimental DRNN is a sequential model (named sequential_1) that is implemented using one LSTM layer along with 3 layers, among which the last layer is a linear layer to shape the target dataset shown in Fig. 3. In real-time noise removal from a signal, the DRNN prepares hidden states that learn the nonstationary pattern (temporal) of noise and process the signal dynamically. This property of the DRNN helps to remove transient noise from the data.

2.2 FCN-DAE

FCN-DAE autoencoder logically works in two phases in completing experiment of denoising and reconstruction. First part, encoder uses input, convolutional and batch normalization layers. Second part uses transposed convolution (interpolation), batch normalization and output layer with activation and pooling layers in both section of model. These all layers are in sequence to make ensure shape of signal after construction. Above mentioned layer serves for a specific purpose to

**Fig. 3** Model Summary Deep Recurrent Neural Network

denoising and reconstruction, where convolutional layer extracts feature, capturing local dependencies and filtering out noise without losing ECG peak, batch normalization layer manages and stabilize training, improve convergence speed, and prevent internal covariate shifts. Feature down sampling is required to enhance noise suppression which retaining signal component which is helped by max-pooling layer. Overall encoder layered sequence, compresses the input signal into a lower-dimensional, reducing redundancy and isolating clean signal features. At this phase, Jacobian function applied to ensure compress representation of ECG signal into low dimensional latent representation with an assurance of small perturbation of noise do not cause large variation. The next consecutive phase is decoder which reconstructs the denoised ECG from this compressed latent space, restoring key cardiac waveforms with minimal distortion. Skip connections help retain spatial details lost in deeper layers, improving reconstruction accuracy. Finally, loss functions like Mean Squared Error (MSE) guide the network to minimize signal reconstruction errors. Layering structure helps to ensures efficient ECG denoising, making FCN-DAE a powerful tool for preprocessing ECG signals.

Moreover, different layers used in the autoencoder configuration affect their ability to learn. A deeper network is susceptible to overfitting, but it can capture complex features. However, increasing the number of features complicates learning. Considering the related concerns when building the model discussed above, a balanced depth in the model, features and proper activation function are chosen and compared against the number of configurations toward fixing layers, capacity and activation in the proposed experiment. After finalization, the FCN-DAE, the input signal undergoes encoding through a series of convolutional layers (6 layers), followed by decoding via transposed convolutions (7 layers) to reconstruct the original signal. The model parameters, including filter size (128), number of filters (7), and activation functions (ReLU), are important in shaping the network's ability to capture and remove noise effectively.

Moreover, increasing the number of layers leads to overfitting and increased computational costs. The reason behind choosing fewer layers in the DRNN is that the long-term dependencies were adequately captured with fewer layers and more layers raised gradient issues. In FCN-DAE, comparatively more layers are used because a reduced number of layers forces less feature extraction in the training model. However, more feature extraction helps in learning against complex signal patterns. During training, the network learns to minimize the discrepancy between the original and reconstructed signals, which are typically quantified via parameters such as SSD, RSME and MAD. Through iterations of forward passes, loss computations, and backpropagation, the parameters of the FCN-DAE are optimized to enhance the denoising performance, ultimately producing the cleaner ECG signal model configuration shown in Fig. 4.

2.3 Dataset

The signal data were collected from the PhysioNet web repository, which is publicly available at <https://physionet.org/content/qtdb/1.0.0/>. The collected QT database is available with a sample frequency of 360 with standard signal data despite the noise data being available in the MIT-BIH noise stress database (NSTDB). The database available in the repository is available in the web from file.atr., hea., dat. and. xws formats and converted the numerical dataset available in the WFDB package in the python library. This QT database has 100, fifteen-minute two-lead ECG recording data, and in the experiment, NSTDB helps in preparing noisy data using a 12-half-hour recording and 3-half-hour noise ambulatory recording [16]. For data preparation, 3 standard noises (baseline wander [bw], muscle[.ma] and electrode motion [.em]) are assembled from the recordings by selecting intervals in two respective noise channels for each type. The training and testing sample distributions are 72002 and 13316 records on 512 sample data points, respectively. The test data have been categorized into 6 types of labels for mention in QT database communication on the basis of the severity of the arrhythmia type.

2.4 Algorithm

The proposed FCN model is trained through ECG signal features in a supervised manner and denoises highly noisy signals via paired information from a clean and noisy ECG signal dataset. The proposed method can be defined as follows: for a given paired training dataset (original and noisy) $(\hat{\mathbf{x}}, \mathbf{x})_1 \dots (\hat{\mathbf{x}}, \mathbf{x})_N$, where $\hat{\mathbf{x}}$ represents the noisy signal vector of the clean signal vector $\mathbf{x}$. For this given pair, the aim is to train the right parameter values via FCN weights so that the network can map the noisy signal $\hat{\mathbf{x}}$ to the clean signal $\mathbf{x}$. With these optimized parameters, the network may be reused to denoise all referenced ECG signals, including those not available in the training database. Fig. 4 shows the flow of the proposed model involving training and inference stages. The proposed model f_{encoder} is composed of 6 dilated convolutional layers along with batch normalization and lambda. This encoder works to downscale the noisy signal $\hat{\mathbf{x}}$ to the compressed lower dimension vector

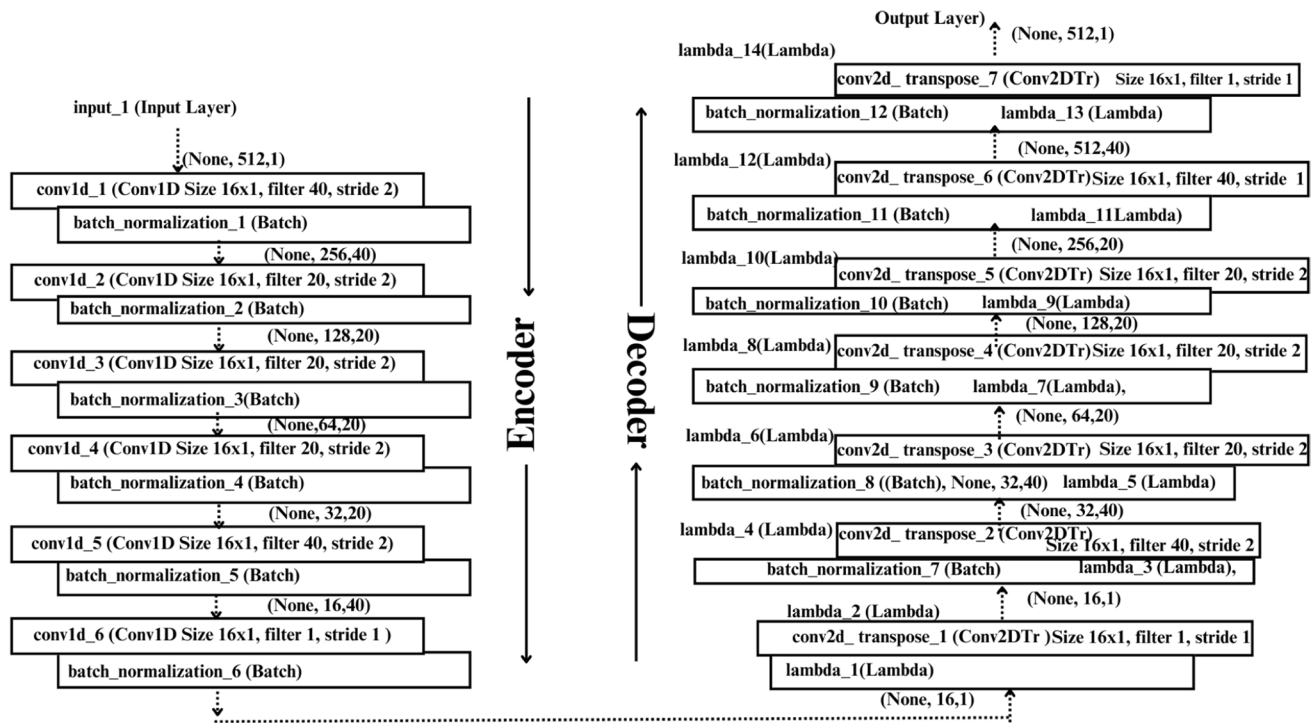


Fig. 4 Model summary of the proposed autoencoder model FCN-DAE

$\hat{\mathbf{z}}$. The encoder performs mapping of $\hat{\mathbf{x}}$ and $\hat{\mathbf{z}}$ to cover only the best features. The decoder upscales by involving 7 layers for upscaling $\hat{\mathbf{z}}$ to $\mathbf{x} \sim$ with the same dimensionality. The discriminator finds the difference to tune λ for optimization of the model. The second part in Fig. 4 shows the overall final filter by these autoencoder and decoder functions. The overall model implementation in the Google Colab environment is shown in Fig. 4.

Correcting for significant changes in the prediction $\hat{\mathbf{x}}$ relative to tiny changes in the input $\mathbf{x}$ is the aim of Jacobian regularization. By doing so, the network becomes more resilient to noise-polluted input data. The partial derivatives of $\hat{\mathbf{x}}$ with respect to $\mathbf{x}$ are contained in the Jacobian $\mathbf{J}$, which encodes the change in prediction with respect to the input.

$$\mathbf{J} = \begin{bmatrix} \frac{\partial \hat{\mathbf{x}}_1}{\partial \mathbf{x}_1} & \cdots & \frac{\partial \hat{\mathbf{x}}_1}{\partial \mathbf{x}_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial \hat{\mathbf{x}}_m}{\partial \mathbf{x}_1} & \cdots & \frac{\partial \hat{\mathbf{x}}_m}{\partial \mathbf{x}_n} \end{bmatrix} \quad (1)$$

The Frobenius norm of the Jacobian matrix aims to minimize the calculated loss when training the network, but it is expensive to draw m backward passes. where sample random noise $\mathbf{v} \mathbf{k} \sim \mathbf{N}(\mathbf{0}, \mathbf{I}_m)$ is a draw from the standard normal distribution and $\mathbf{I}$ is the m -by- m identity matrix that enforces loss. The classical DAE loss force model is used to reconstruct the input signal accurately, and Jacobian regularization measures the sensitivity of the reconstructed signal with respect to small changes in noisy data. The Jacobian matrix maintains local information preservation while denoising the reconstructed signals.

$$\text{Total Loss} = \text{Reconstruction Loss} + \text{Regularization Term}$$

$$\text{Reconstruction through DAE} = \mathcal{L}_{\text{DAE}} = \|\mathbf{X} - \hat{\mathbf{X}}\|^2 \quad (2)$$

$$\text{Regularization Term} = \mathcal{L}_{\text{Jacobian}} = \|\mathbf{J}_{\hat{\mathbf{X}}} - \mathbf{I}\|_F^2 \quad (3)$$

$$\text{TotalLoss} = \mathcal{L} = \mathcal{L}_{DAE} + \lambda \cdot \mathcal{L}_{\text{Jacobian}} \quad (4)$$

Here, $J_{\hat{X}}$ is the Jacobian matrix of the reconstructed output $\hat{X}$ with respect to the input X , and I is the identity matrix. $\|\cdot\|_F$ denotes the Frobenius norm. The parameter λ controls the trade-off between these two objectives. Tuning to λ affects the importance of noise reduction versus local information preservation. Higher values of λ prioritize preserving local information, whereas lower values prioritize noise reduction.

2.5 Performance measures

The performance measure is an important part of any experiment in artificial intelligence and deep learning. These performance measures help guide us in performance optimization and reach a satisfactory level in the experimental model through tuning parameters in the training model. Common and widely used statistical measures for loss include the mean absolute deviation (MAD), also known as the mean absolute difference, and the sum of squared difference (SSD), also known as the sum of squared error (SSE) and root mean squared error (RSME), which are given in Eqs. 1, 2, and 3. All the measures are calculated over the predicted values and observed values from the experiments over the data points.

$$SSD = \sum_{i=1}^N (x_i - \hat{x}_i)^2 \quad (5)$$

$$MAD = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i| \quad (6)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2} \quad (7)$$

2.6 Implementation

The implementation setup and procedures used to investigate the performance of the proposed work. Initially, a comprehensive dataset of original and noisy signals was collected to perform the experiment. The noisy signal was prepared by contamination of standard multichannel noise, such as baseline wander (.bw), muscle artifact (MA), and powerline interference, and electrode contact noise was collected. The FCN-DAE architecture is subsequently implemented and trained on the noisy ECG dataset, aiming to confine and preserve local information while effectively removing noise. The proposed FCN-DAE model is constructed with convolutional layers for feature extraction and transposed convolutional layers for signal reconstruction, with appropriate activation functions and optimization techniques employed. The training process involves minimizing the discrepancy between the original and denoised ECG signals via suitable loss functions, such as the mean squared error (MSE) or variations tailored to handle multichannel noise. Stability and robustness analysis is important concern to understand effectiveness of Jacobin function. Perturbation inputs are an important concern before training and testing, signal components are noisy or distorted, not appropriate in signal analysis too in medical field. To increase the stability of models, several regularization techniques have been put forth. This objective is achieved for support vector machines and linear classifiers using an L^2 regularization that minimizes overfitting to the training data and maximizes classification margins. Neural networks have also made extensive use of this regularization strategy, which has been demonstrated to enhance generalization [Zhang et al., 2018]. However, it remains unclear whether or not L^2 regularization increases classification margins and stability of a network, especially for deep architectures with intertwining nonlinearity. For a classification model f is a function that uses X input which is a signal vector and every component is defined by x . Perturbed input can be defined as $\tilde{x} = x + \epsilon$, where ϵ small vector in similar dimension as input, produce corresponding shift in output defined by.

$$\tilde{z}_c = f_c(x + \epsilon) = f_c(x) + \sum_{i=1}^I \epsilon_i \cdot \frac{\partial f_c}{\partial x_i}(x) + \mathcal{O}(\epsilon^2) = z_c + \sum_{i=1}^I J_{c,i}(x) \epsilon_i + \mathcal{O}(\epsilon^2) \quad (8)$$

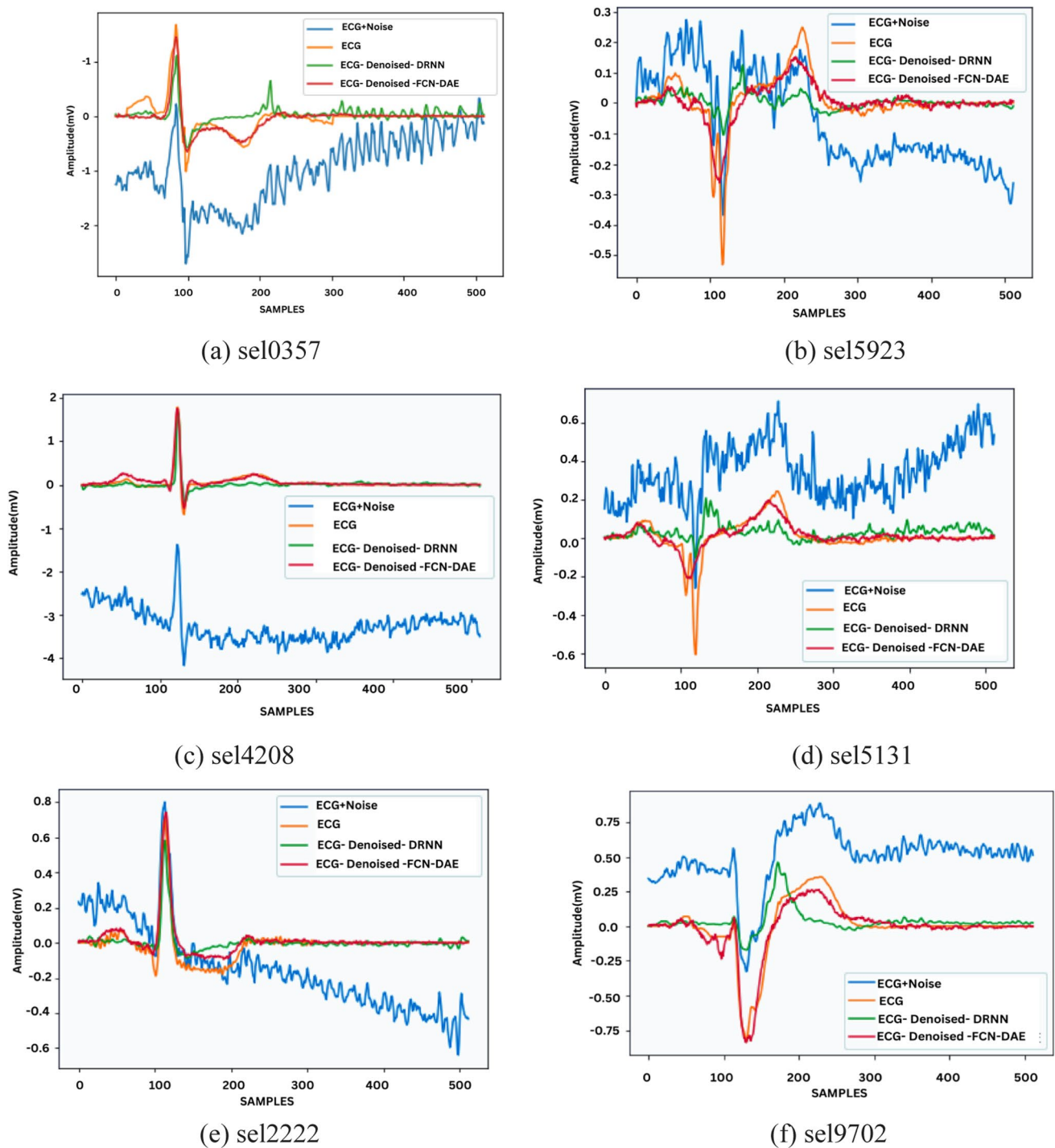


Fig. 5 a–f ECG signal plot of standard, noise-included, and denoised (DRNN, proposed FCN-DAE) patterns for 6 types of test cases selected [0357,5923,4208,5131,2222 and 9702] from the test dataset mentioned in the QT database for different heart disorders

The above equation is a Taylor-expanded with respect to the input perturbation ϵ and in the third equality the defines Jacobian matrix as.

$$J_{c,i}(\mathbf{x}) = \frac{\partial \mathbf{f}_c}{\partial \mathbf{x}_i}(\mathbf{x}) \quad (9)$$

Table 3 Comparison of performance

Method/Model	SSD (max/min) $\times 10^{-2}$	MAD (max/min)	RMSE (max/min)
LSTM Encoder	13.630 (12.568)	0.750 (0.459)	3.472 (2.561)
Stacked Autoencoder	12.88 (12.65)	0.658 (0.457)	2.989 (2.356)
Attention Mechanism	10.75 (11.506)	0.466 (0.398)	3.129 (1.869)
Transformer encoder	9.56 (9.2)	0.526 (0.458)	1.956 (1.889)
DRNN	12.630 (11.486)	0.689 (0.384)	3.047 (1.483)
Proposed Model (Without Jacobian)	7.252 (4.736)	0.429 (0.288)	1.859 (1.131)
Proposed Model (With Jacobian)	7.038 (4.03)	0.426 (0.248)	1.678 (1.112)

Each element z_c is associated with likelihood of input, for small perturbations $\in$ the higher-order terms can be neglected and the stability can be seen. A simplest process to minimize this instability through reducing the magnitude for each component of the Jacobian matrix, and realization possible through square of the Frobenius norm (given below).

$$\|J(x)\|_F^2 = \left\{ \sum_{i,c} [J_{c,i}(x)]^2 \right\} \quad (10)$$

3 Result

The proposed work discussed in Sect. 2 focuses on the background of the experiment. In the first step, the DRNN model was implemented as a sequential model, and the ReLU optimization function was used. In the second step, the FCN-DAE was implemented and tested with a Jacobian objective function and without a Jacobian objective function. The results are collected and presented as visual patterns and compared across different error attributes in tabular format. The results of the experiments with the DRNN and proposed model (FCN- DAE) were collected and compared with normal ECGs and noise-embedded ECG signals from the NSTDB dataset. A comparison of the loss SSD, MAD and RSME results across 13362 test cases is shown in Table 3. The shapes of the ECGs were compared through random picked test result ECG data on plotted graph ECG graph generation. The graphical results compare the ECG signal waves and the visualization of noisy and denoised signals. The results of the experiment generated and plotted on a 0.2–2 mV amplitude of signal (noisy and denoised) on 512 sample data points. Figure 5a–f shows the standard ECG, ECG + noise, DRNN denoise d and proposed FCN-DAE denoised signals over the selected wave form to compare local information preservation from other types. Six test cases taken from the QT dataset are used as test cases, where comparisons are shown for 0357, 5923, 4208, 5131, 222 and 9702 signals generated in the QT database. From Figure 5a–f, it is clear that the proposed implemented autoencoder with a customized layer and objective function design helps reduce loss and improve local information preservation. In every graph, the denoised signal from the proposed model looks better in shape and appropriate segment information, details given in Table 4.

The statistical and visual results of the proposed method are better than those of the DRNN. The proposed method improves performance by reducing the loss in terms of SSD, RMSE and MAD up to 4.736, 0.288, and 1.131 (without Jacobian) and 4.03, 0.248, and 0.1112 (with Jacobian), respectively. The DRNN-based pattern after denoising looks fine but for local information identification because of muscle or electromagnetic effects is not possible with the DRNN. Figure 6 shows how much local information is near the clean signal after denoising over 1000 data points of a noisy signal. The pattern of the denoised signal is much closer to the clear signal pattern, where different peaks, segments

Table 4 Graph output cases and description

Graph output Cases	Description
ECG	ECG collected from Dataset
ECG + Noise	Signal Encoded with 3, noise in NSTDB
ECG Denoise Using DRNN	Denoised signal through Deep Recurrent Neural Network
ECG Denoise Using Proposed Model	Fully convolutional neural network for auto encoding and denoising

Fig. 6 Comparison of noisy signals and denoised signals from clean signals via the proposed method on a long test sample of 1000 data points

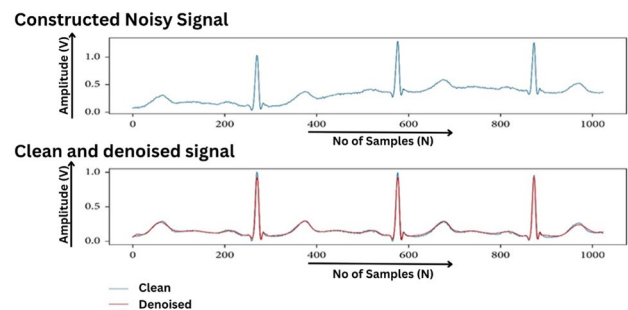


Fig. 7 Effect of the Jacobian function in regularization over a small (50 epochs)

Before training:

accuracy: $827/10000=0.083$

supervised loss: 2.675

Jacobian loss: 3.656

Total loss: 3.041

After 50 epochs of training:

accuracy: $9702/10000=0.9702$

supervised loss: 0.027

Jacobian loss: 0.977

total loss: 0.125

and intervals can be clearly identified via usual diagnostic methods. In specific patients, denoised ECGs help identify heart abnormalities, medical conditions and severity. The individual effect of the Jacobian function in the regularization of the model was checked before training and after 50 epochs and was found to be effective up to 97.02 in terms of accuracy, as shown in Fig. 7. The Jacobian function is part of the optimization and regularization involved, and the results are noted for the error matrix in the standard 50 epochs. Improvements (in %) over the standard error matrix SSD, MAD and RMSE were noted against the effect of the Jacobian matrix shown in Table 5, and significant improvements were found. In addition to the statistical comparisons mentioned in referenced tables, visual representations of the results are also given. Referenced six different labeled query signals and a comparison of denoised signals from the QT-Signal dataset clearly shows that after denoising, the signal shape and local information, such as the amplitude and frequency, are preserved. The comparison line graph shown in Fig. 8 focuses on comparing the error matrix of the proposed model and previous models, such as the LSTM, attention mechanism, DRNN, stacked encoder and transformer encoder. Figure 8 shows that overall, correction of any noisy signal is possible through the proposed

Fig. 8 Performance comparison graph

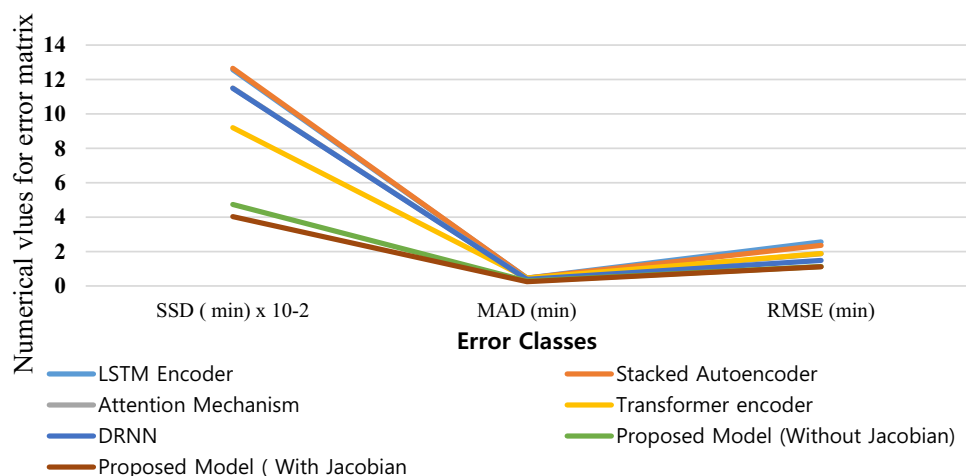


Table 5 Effect of the Jacobian function

Noise Type	Metric	Without Jacobian	With Jacobian	Improvement (%)
Mixed Noise	SSD (mV ²)	0.047	0.040	14.89%
	MAD (mV)	0.288	0.248	13.88%
	RMSE (mV)	0.013	0.011	15.38%

model with accuracy, as shown. From the figures, it is also clear that the % loss for local information is too low such that the structure can be easily reconstructed to understand the actual meaning of the ECG signal for analysis.

This work suggests a method that reducing noise can help preserve the morphology of signals (e.g., P, QRS, and T waves) as much as possible, which is necessary for diagnosis by a doctor or computer-based diagnostic system. To acknowledge the signal-to-noise ratio (SNR), which is an important parameter for model evaluation, the initial noise level in the decibel is considered in noisy ECG signals under experimentation, and the results are collected for denoised signals. Table 6 shows the initial levels of noise and noise labels in the denoised signal in the SNR (resulting from the DRNN and FCN-DAE) for the given record IDs: sel0357, 5923, 4208, 5131, 2222 and 9702. Compared with that of the DRNN, the SNR level of the proposed method significantly improved. With respect to suggestive guidelines for ECG diagnosis, the classification task for different diseases related to the heart requires future work, which will be carried out using denoised signal data. Future work will explicitly validate ECG diagnosis by integrating diagnostic tasks into the evaluation pipeline. Effect of Jacobin function for stabilizing input has been seen in experimental results in terms of test accuracy over given taken sample has been shown in Table 7. Jacobin regularizer shows best results in testing accuracy which proves robustness of proposed model for denoising ECG signal (noisy/distorted).

4 Discussion

The proposed deep learning model introduces a novel FCN-based denoising autoencoder specifically optimized for multichannel ECG signals. A key point of novelty is the integration of a Jacobian regularization that helps to improve feature sensitivity and preserve critical morphology in denoised signals. This work outperforms on conventional methods like DRNN by reducing different types of ECG noise and maintaining computational efficiency along with clinical interpretability. The objective found in Sect. 1, was to prepare and test a deep learning model that ensures minimum loss of local information in the ECG signal during the denoising process. In the claimed work, the FCN-DAE model is customized and supported by a Jacobian optimization function that highlights local and global information in ECG denoising. Although the Jacobian matrix is a conventional and well-known method, it helps to push decision boundaries of related parameters

Table 6 Improvement in the SNR by the implemented model

Record ID	Baseline SNR (dB)	Output SNR—DRNN (dB)	Output SNR—FCN-DAE (dB)	Improvement in SNR (DRNN)	Improvement in SNR (FCN-DAE)
sel0357	5.2	12.5	15.8	7.3	10.6
sel5923	3.8	11.2	13.9	7.4	10.1
sel4208	4.5	10.8	14.3	6.3	9.8
sel5131	2.9	9.7	12.4	6.8	9.5
sel2222	3.2	10.1	13.7	6.9	10.5
sel9702	4	11.3	14.5	7.3	10.5

Table 7 Effect of Jacobian regularizer showing substantially reduces the norm of the Jacobian while retaining test accuracy

Regularizer	Test accuracy	
	Proposed model	$\ J\ _F$ (↓)
No regularization	95.6 ± 0.1	32.9 ± 3.5
L2	96.9 ± 0.1	4.6 ± 0.2
Dropout	96.9 ± 0.1	3.5 ± 0.2
Jacobian	97.02 ± 0.1	1.1 ± 0.1

and provide a stable learning model for application. Each regularization method attempts to constrain the learning data and Jacobian matrix. This effectively reduces the value of the Frobenius norm of the Jacobian matrix. This improves the generalizability of an unseen test dataset and constraint boundary for experiments. The experimental results from the claimed article show that Jacobian regularization provides significant robustness against the corruption of input data of both random and adversarial types. The experimental evaluation of the given articles reveals the effectiveness of Jacobian regularization, which substantially reduces the norm and retains test accuracy for new custom-built deep learning models. This article applies modification in FCN-DAE, addresses challenges in multichannel ECG denoising via Jacobian regularization and outperforms conventional approaches in terms of signal quality and clinical applications. The modified strategy addresses previously unresolved issues in model training, such as randomization issues in learning patterns, inter-channel correlations, and signal morphology. The denoised signal helps preserve more detailed information via convolution throughout the fully connected structure. The proposed method also ensures the consistency of the information distributions in the original signal and the signal after denoising, effectively maintaining the global information showed in Moreover, FCNs are flexible tools for handling sequential data, including time series that provide feature time series information. The experimental results of the proposed model are good for the SSD, MAD, and RMSE matrices. The analysis of denoised ECGs shows that the proposed approach outperforms the other models in terms of local preservation and noise reduction. This local information helps medical professionals find clearer patterns in ECGs in real-time systems. However, a significant amount of computation is needed for our solution to involve the Jacobian function for regularization. Lightweight denoising models, which can be employed for portable wearable devices, are another crucial topic for future research. Furthermore, numerous other active and efficient ECG-denoising models exist that require validation on the basis of performance parameters. Optimization in deep learning-based autoencoders requires a new fine-tuning method to develop or find alternatives according to noise.

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Declarations

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References

1. Abel JDK, Dhanalakshmi S, Kumar R. A comprehensive survey on signal processing and machine learning techniques for noninvasive fetal ECG extraction. *Multimed Tools Appl.* 2022;82(1):1373–400.
2. Andreotti F, Behar J, Zaunseder S, Oster J, Clifford GD. An open-source framework for stress-testing noninvasive fetal ECG extraction algorithms. *Physiol Meas.* 2016;37(5):627–48.
3. Andreotti F, Riedl M, Himmelsbach T, Wedekind D, Wessel N, Stepan H, Schmieder C, Jank A, Malberg H, Zaunseder S. Robust fetal ECG extraction and detection from abdominal leads. *Physiol Meas.* 2014;35(8):1551–67.

4. Andreotti F, Gräber F, Malberg H, Zaunseder S. Noninvasive fetal ECG signal quality assessment for multichannel heart rate estimation. *IEEE Trans Biomed Eng.* 2017;64(12):2793–802.
5. Arsene CT C, Hankins R and Yin H, 2019, Deep Learning Models for Denoising ECG Signals, *27th European Signal Processing Conference (EUSIPCO), A Coruna, Spain*: 1–5.
6. Azarang A, Kehtarnavaz N. A review of multiobjective deep learning speech denoising methods. *Speech Commun.* 2020;122:1–10.
7. Biswal K, Swain S, Tripathy MC. A high-Q fractional-order twin-T notch filter for low-frequency applications. *Sādhanā.* 2023;48:38.
8. Bouny L El, Khalil M and Adib A, 2021, Convolutional Denoising Auto Encoder Based AWGN Removal from ECG Signal, *International Conference on INnovations in Intelligent SysTems and Applications (INISTA)*, Kocaeli, Turkey:1–6.
9. Chatterjee S, Thakur RS, Yadav RN, Gupta L, Raghuvanshi DK. Review of noise removal techniques in ECG signals. *IET Signal Proc.* 2020;14(9):569–90.
10. Craik A, He Y, Contreras-Vidal JL. Deep learning for electroencephalogram (EEG) classification tasks: a review. *J Neural Eng.* 2019;16(3):031001.
11. Chen FQ, Wu Y, Zhao GD, Zhang JM, Zhu M, Bai J. Contractive denoising AutoEncoder. *Intell Comput Theory, Lect Notes Comput Sci.* 2014;8588:776–81.
12. Chen S, Guo W. AutoEncoders in deep learning-a review with new perspectives. *Mathematics.* 2023;11(8):1777.
13. Chiang HT, Hsieh YY, Fu SW, Hung KH, Tsao Y, Chien SY. Noise reduction in ECG signals using fully convolutional denoising autoencoders. *IEEE Access.* 2019;7:60806–13.
14. Chiang, H T, Hung K H, Fu S W, Kuo H C, Tsai M H and Tsao Y, 2023, Study on the Correlation Between Objective Evaluations and Subjective Speech Quality and Intelligibility, *IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), IEEE Xplore*:1–7.
15. Festag S, Spreckelsen C. Medical multivariate time series imputation and forecasting based on a recurrent conditional Wasserstein GAN and attention. *J Biomed Inform.* 2023;139:104320.
16. Fotiadou E, Konopczyński T, Hesser J, Vullings R. End-to-end trained encoder–decoder convolutional neural network for fetal electrocardiogram signal denoising. *Physiol Meas.* 2020;41(1):015005.
17. Goldberger AL, Amaral LAN, Glass L, Hausdorff JM, Ivanov PC, Mark RG, Mietus JE, Moody GB, Peng CK, Stanley HE. PhysioBank PhysioToolkit, and PhysioNet. 2000. *Circulation.* <https://doi.org/10.1161/01.CIR.101.23.e215>.
18. Hinton G, Deng L, Yu D, Dahl G, Mohamed AR, Jaitly N, Senior A, Vanhoucke V, Nguyen P, Sainath T, Kingsbury B. Deep neural networks for acoustic modeling in speech recognition: the shared views of four research groups. *IEEE Signal Process Mag.* 2012;29(6):82–97.
19. Isler Y, Ozturk U, Sayilgan E. A new sample reduction method for decreasing the running time of the k-nearest neighbors algorithm to diagnose patients with congestive heart failure: backward iterative elimination. *Sādhanā.* 2023;48:35.
20. Joo-Chang K, Chung K. Multi-modal stacked denoising autoencoder for handling missing data in healthcare big data. *IEEE Access.* 2020. <https://doi.org/10.1109/ACCESS.2020.2997255>.
21. Lei H, Yang Y. CDAE: a cascade of denoising autoencoders for noise reduction in the clustering of single-particle cryo-EM images. *Front Genet.* 2021;11:627746.
22. Mohan Rao B, Kumar A. Detection of heart arrhythmia based on UCMFB and deep learning technique. *Sādhanā.* 2022;47:252.
23. Hamad M, Alhussainy A, Jasim DA. ECG signal denoising based on deep learning auto encoder and discrete wavelet transform. *Int J Eng Technol.* 2020;9(2):415–23.
24. Martinsen O, Grimnes S. Line patterns in the mosaic electric properties of human skin-a cross correlation study. *IEEE Trans Biomed Eng.* 2003;50(1):114.
25. Patel R, Gireesan K, Baskaran R. Optimal classification of N-back task EEG data by performing effective feature reduction. *Sādhanā.* 2022;47:281.
26. Pinto P, Costa-Santos C, Gonçalves H, Ayres-De-Campos D, Bernardes J. Improvements in fetal heart rate analysis by the removal of maternal-fetal heart rate ambiguities. *BMC Pregnancy Childbirth.* 2015;15(1):301.
27. Qiu C, Li H, Qi C, Li B. Enhancing ECG classification with continuous wavelet transform and multibranch transformer. *Heliyon.* 2024. <https://doi.org/10.1016/j.heliyon.2024.e26147>.
28. Romero FP, Piñol DC, Vázquez-Seisdedos CR. DeepFilter: an ECG baseline wander removal filter using deep learning techniques. *Biomed Signal Process Control.* 2021;70:102992.
29. Samann F, Schanze T. RunDAE model: running denoising autoencoder models for denoising ECG signals. *Comput Biol Med.* 2023;166:107553.
30. Su J, Xu B, Yin H. A survey of deep learning approaches to image restoration. *Neurocomputing.* 2022;487:46–65.
31. Teplitzky BA, McRoberts M, Ghanbari H. Deep learning for comprehensive ECG annotation. *Heart Rhythm.* 2020;17(5):881–8.
32. Tracey BH, Miller EL. Nonlocal means denoising of ECG signals. *IEEE Trans Biomed Eng.* 2012;59(9):2383–6.
33. Ullah H, Bin Heyat MB, Akhtar F, Sumbul MAY, Islam MS, Abbas Z, Pan T, Gao M, Lin Y, Lai D. An end-to-end cardiac arrhythmia recognition method with an effective DenseNet model on imbalanced datasets using ECG signal. *Comput Intell Neurosci.* 2022. <https://doi.org/10.1155/2022/9475162>.
34. Vullings R, van Laar JOEH. Noninvasive fetal electrocardiography for intrapartum cardiotocography. *Front Pediatr.* 2020;8:599049.
35. Xiong P, Wang H, Liu M, Lin F, Hou Z, Liu X. A stacked contractive denoising autoencoder for ECG signal denoising. *Physiol Meas.* 2016;37(12):2214–30.
36. Xiong P, Wang H, Liu M, Zhou S, Hou ZG, Liu X. ECG signal enhancement based on improved denoising autoencoder. *Eng Appl Artif Intell.* 2016;52:194–202.
37. Zhang Q, Fu L, Gu L. A cascaded convolutional neural network for assessing signal quality of dynamic ECG. *Comput Math Methods Med.* 2019. <https://doi.org/10.1155/2019/7095137>.
38. Zhong W, Liao L, Guo X, Wang G. A deep learning approach for fetal QRS complex detection. *Physiol Meas.* 2018;39(4):045004.
39. Zhou Z, Huang K, Qiu Y, Shen H, Ming Z. Morphology extraction of fetal electrocardiogram by slow-fast LSTM network. *Biomed Signal Process Control.* 2021;68:102664.
40. Mvuh FL, Ko C, Bodo B. Multichannel high noise level ECG denoising based on adversarial deep learning. *Sci Rep.* 2024. <https://doi.org/10.1038/s41598-023-50334-7>.

41. Co KT, Rego DM, Lupu EC. Jacobian regularization for mitigating universal adversarial perturbations. In: Farkaš I, Masulli P, Otte S, Wermter S, editors. Lecture notes in computer science. Cham: Springer; 2021. p. 202–13.
42. Hoffman J, Roberts DA, Yaida S. Robust learning with jacobian regularization. ArXiv. 2019. <https://doi.org/10.48555/arXiv.1908.02729>.
43. Liu D, Wu LY, Li B, Boussaid F, Bennamoun M, Xie X, Liang C. Jacobian norm with selective input gradient regularization for interpretable adversarial defense. Pattern Recogn. 2023;145:109902. <https://doi.org/10.1016/j.patcog.2023.109902>.

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